INTRODUCTION TO STOCHASTIC PROCESSES

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Fifth Semester

Contents

Chapter 1

RANDOM WALKS

July 22nd.

Quizzes will account for 20% of the final grade, and assignments for 5%. The mid-term will account for 25% of the final grade, and the final exam will account for 50%. The primary reference for random walks is H. Follner.

1.1 The Setup

We wish to concretify the idea of a random walk on the integers $\mathbb{Z}$. Ω is out sample space, with $0 < |\Omega| < \infty$ (we shall later look at the case when Ω is countably infinite). $\mathcal{F}$ is the event space, and is exactly $\mathcal{P}(\Omega)$, the power set of Ω. The intuition of a random walk is as follows; stand at the origin. At each step, flip a fair coin. If the coin comes up heads, take a step to the right; if tails, take a step to the left. Repeat this process N times. S_n here then denotes the position of the random walk after n steps, for $0 \leq n \leq N$.

For a fixed $N \in \mathbb{N}$, we let $\Omega_N = \{\pm 1\}^N$, and $\mathcal{F}_N = \mathcal{P}(\Omega_N)$. For $1 \leq k \leq N$, define the random variable $X_k : \Omega_N \to \{\pm 1\}$ as $X_k(\omega) = \omega_k$, where $\omega = (\omega_1, \omega_2, \ldots, \omega_N) \in \Omega_N$; this models the step taken by the random walk at step k. Further, define $S_0 = 0$, and for $1 \leq n \leq N$, define $S_n : \Omega_N \to \mathbb{Z}$ as $S_n(\omega) = \sum_{k=1}^n X_k(\omega)$; this models the position of the random walk after n steps. Finally, define the probability measure $\mathbb{P}_N$ on $(\Omega_N, \mathcal{F}_N)$ as $\mathbb{P}_N(A) = \frac{|A|}{2^N}$ for $A \in \mathcal{F}_N$. This models the fair coin flip at each step of the random walk.

Definition 1.1. The sequence of random variables $\{S_n\}_{n=0}^N$ on $(\Omega_N, \mathcal{F}_N, \mathbb{P}_N \equiv \mathbb{P})$, as prescribed above, is called the *simple symmetric random walk* of length N on $\mathbb{Z}$, starting at the origin.

Given $1 \leq k_1 < k_2 < \cdots < k_n \leq N$, and $x_i \in \{\pm 1\}$,

$$\mathbb{P}(X_{k_1} = x_1, \ldots, X_{k_n} = x_n) = \frac{|\{\omega \in \Omega_N \mid X_{k_i}(\omega) = x_i, 1 \leq i \leq n\}|}{2^N} = \frac{2^{N-n}}{2^N} = 2^{-n}. \tag{1.1}$$

That is, fixing n steps of the walk, the probability of any particular sequence of steps is 2^{-n}. Using this, we can show that for any $1 \leq k \leq N$, $\mathbb{P}(X_k = 1) = \mathbb{P}(X_k = -1) = \frac{1}{2}$. Moreover, we can extract that

$$\mathbb{P}(X_k = x_k, X_l = x_l) = 2^{-2} = \mathbb{P}(X_k = x_k)\mathbb{P}(X_l = x_l), \tag{1.2}$$

showing independence of subsequent steps; the model seems like a reasonable fit for our intuition of a random walk.

Example 1.2. The simple symmetric random walk has independent increments; that is, given $0 < k_1 < k_2 < N$, the random variables $S_k - S_0$, $S_{k_2} - S_{k_1}$, and $S_N - S_{k_2}$ are mutually independent. This is because the increments are functions of disjoint sets of independent random variables X_k, and hence are independent themselves. Similarly, one may concretely show that for $1 \leq k < m \leq N$, the distribution of $S_m - S_k$ is the same as that of S_{m-k}.

The following properties of the simple symmetric random walk are immediate consequences of the above definition and example.

Suppose we have $\mathbb{P}(S_{n-1} = a_{n-1}, S_{n-2} = a_{n-2}, \ldots, S_0 = 0) > 0$. Then, we are free to compute the conditional probability $\mathbb{P}(S_n = a_n \mid S_{n-1} = a_{n-1}, S_{n-2} = a_{n-2}, \ldots, S_0 = 0)$. One may verify that

$$\mathbb{P}(S_n = a_n \mid S_{n-1} = a_{n-1}, S_{n-2} = a_{n-2}, \ldots, S_0 = 0) = \mathbb{P}(S_n = a_n \mid S_{n-1} = a_{n-1}), \tag{1.3}$$

for all $a_1, \ldots, a_n \in \mathbb{Z}$ such that $\mathbb{P}(S_{n-1} = a_{n-1}, S_{n-2} = a_{n-2}, \ldots, S_0 = 0) > 0$. This is called the *Markov property* of the simple symmetric random walk. In words, this property states that given the present state of the walk, the future evolution of the walk is independent of its past history.

Now suppose $\mathbb{P}(S_k = a) > 0$ for some $k \geq 1, a \in \mathbb{Z}$, and let $m > k, b \in \mathbb{Z}$. Then, one can simply compute that $\mathbb{P}(S_m = b \mid S_k = a) = \mathbb{P}(S_{m-k} = b - a)$.

The next question we may ask is, on an average, where the random walk is located after n steps, for $1 \leq n \leq N$. We can compute the expected value of S_n as follows:

$$\mathbb{E}[S_n] = \mathbb{E}\left[\sum_{k=1}^{n} X_k + 0\right] = \sum_{k=1}^{n} \mathbb{E}[X_k] = \sum_{k=1}^{n} 0 = 0. \tag{1.4}$$

Thus, to get a better idea of the location of the random walk after n steps, we compute the variance of S_n:

$$\operatorname{Var}(S_n) = \operatorname{Var}\left(\sum_{k=1}^{n} X_k + 0\right) = \sum_{k=1}^{n} \operatorname{Var}(X_k) = \sum_{k=1}^{n} 1 = n. \tag{1.5}$$

So, via the central limit theorem, we can approximate for large n as

$$\mathbb{P}\left(-3 \leq \frac{S_n}{\sqrt{n}} \leq 3\right) \approx \Phi(3) - \Phi(-3) \tag{1.6}$$

where Φ is the cumulative distribution function of the standard normal distribution.

We now claim that

$$\mathbb{P}(S_{2n} = 0) = \binom{2n}{n} 2^{-2n} = \mathbb{P}(S_{2n-1} = 1) = \mathbb{P}(S_{2n-1} = -1). \tag{1.7}$$

To this end, we note that for $S_n = x$, we must have taken $\frac{n+x}{2}$ steps to the right, and $\frac{n-x}{2}$ steps to the left. Thus, we have

$$\mathbb{P}(S_n = x) = \frac{|\{\omega \in \Omega_N \mid S_n = x\}|}{2^N} = \binom{n}{\frac{n+x}{2}} \frac{2^{N-n}}{2^N} = \binom{n}{\frac{n+x}{2}} 2^{-n}. \tag{1.8}$$

The binomial coefficient attains its maximum at $x = 0$ for even n, and at $x = \pm 1$ for odd n. Thus, we have shown the above required claim. We can further approximate this via *Stirling's formula*, which stats $k! \sim \sqrt{2\pi k} k^k e^{-k}$ for large k. For now, we shall treat this as an equality. Thus, for large n,

$$\mathbb{P}(S_{2n} = 0) = \binom{2n}{n} 2^{-2n} \sim \frac{\sqrt{4\pi n}(2n)^{2n} e^{-2n}}{2\pi n n^{2n} e^{-2n}} 2^{-2n} = \frac{1}{\sqrt{\pi n}}. \tag{1.9}$$

We now show that given any interval $[a, b] \subsetneq \mathbb{R}$, the probability that the random walk stays in this interval is zero as $n \to \infty$. To this end, note that either $\mathbb{P}(S_n = x) \leq \mathbb{P}(S_n = 0)$ or $\mathbb{P}(S_n = x) \leq \mathbb{P}(S_n = 1)$ holds for any given n; moreover, both of the quantities on the right-hand side are bounded by $\frac{c}{\sqrt{n}}$ for an appropriate constant $c > 0$. Thus, we have

$$0 \leq \mathbb{P}(a \leq S_n \leq b) = \sum_{x=a}^{b} \mathbb{P}(S_n = x) \leq \frac{c(b - a + 1)}{\sqrt{n}} \to 0 \text{ as } n \to \infty. \tag{1.10}$$

1.2 Stopping Times

July 29th.

We can interpret S_n as the amount of capital earned by a player until time n. Earlier, we saw that the expected 'amount of capital' at n is $\mathbb{E}[S_n] = 0$, for $0 \leq n \leq N$. We ask if there is a favourite (random) time at which the player should stop playing, so as to maximize the expected amount of capital earned. We assume that there is no insider trading, and that the player's decision to stop playing at time n is based on the information available until time n.

Definition 1.3. A subset $A \subseteq \Omega$ is termed an *observable event* by time n if it can be written as a union of basic events of the form

$$\{\omega \in \Omega \mid \omega_1 = o_1, \ldots, \omega_n = o_n\}, \quad o_1, \ldots, o_n \in \{\pm 1\}. \tag{1.11}$$

That is, A can be determined by observing the outcomes of the first n coin tosses.

We shall use $\mathcal{A}_n$ to denote the collection of all observable events by time n. One can show that

$$\{\emptyset, \Omega\} = \mathcal{A}_0 \subseteq \mathcal{A}_1 \subseteq \cdots \subseteq \mathcal{A}_N = \mathcal{F}. \tag{1.12}$$

Definition 1.4. A map $T : \Omega \to \{0, 1, \ldots, N\} \cup \{\infty\}$ is called a *stopping time* if $\{T = n\} \in \mathcal{A}_n$ for all $0 \leq n \leq N$.

As an example, pick $a \in \mathbb{Z}$. We can then define $\sigma_a : \Omega \to \{0, 1, 2, \ldots, N\} \cup \{\infty\}$ as $\sigma_a(\omega) = \min\{n \geq 0 \mid S_n(\omega) = a\}$, with the convention that $\min \emptyset = \infty$. This is, simply put, the first time the random walk hits the point a. The reader may show that σ_a is indeed a stopping time. On the other hand, the map $\tau : \Omega \to \{0, 1, 2, \ldots, N\} \cup \{\infty\}$ defined as $\tau(\omega) = \min\{n \geq 0 \mid S_n(\omega) = S_N(\omega)\}$ is not a stopping time; this is because we cannot determine if the random walk has hit its final position at time n without knowing the outcome of the coin tosses after time n.

Theorem 1.5. *For any stopping time $T : \Omega \to \{0, 1, \ldots, N\}$, we have $\mathbb{E}[S_T] = 0$.*

That is, there is an impossibility of a favourite stopping time. We shall adopt the notation $S_T(\omega) := S_{T(\omega)}(\omega)$, the outcome of the trajectory ω at the stopping time $T(\omega)$. This leads us to discuss the impossibillity of profitable games.

Let $V_1, \ldots, V_N : \Omega \to \mathbb{R}$ be random variables such that $\{V_k = c\} \in \mathcal{A}_{k-1}$ for all $c \in \mathbb{R}$, $k = 1, \ldots, N$. V_k here represents the amount of capital a player bets at time k. The result at the k-th round of the game is then $V_k X_k$. This is known as a game system.

Theorem 1.6. *For any $V_1, \ldots, V_N$ which are a game system as above, let $S_n^V = \sum_{k=1}^n V_k X_k$ for $1 \leq n \leq N$. Then, $\mathbb{E}[S_N^V] = 0$.*

Proof. We have $\mathbb{E}[S_N^V] = \sum_{k=1}^N \mathbb{E}[V_k X_k]$. Let us restrict our range as $\mathrm{range}(V_k) = \{c_1, \ldots, c_M\}$ for some $M \in \mathbb{N}$. Then, we have

$$\mathbb{E}[V_k X_k] = \sum_{i=1}^M c_i \, \mathbb{E}[\mathbf{1}_{\{V_k = c_i\}} X_k] \tag{1.13}$$

Note that, by definition, $\{V_k = c_i\} \in \mathcal{A}_{k-1}$, and hence is independent of X_k. Thus, we have

$$\mathbb{E}[V_k X_k] = \sum_{i=1}^M c_i \, \mathbb{E}[\mathbf{1}_{\{V_k = c_i\}}] \, \mathbb{E}[X_k] = 0. \tag{1.14}$$

■

We shall use this theorem to prove the theorem before it.

Proof of impossibility of a favourite stopping time. We have

$$S_T = \sum_{k=1}^{T(\omega)} X_k = \sum_{k=1}^N \mathbf{1}_{\{T \geq k\}} X_k. \tag{1.15}$$

Now, for $1 \leq k \leq N$, set $V_k : \Omega \to \{0, 1\}$ as $V_k(\omega) = \mathbf{1}_{\{T(\omega) \geq k\}}$. Then $\{V_k = 1\} = \{T \geq k\} = \{T < k\}^c = \left(\bigcup_{m=0}^{k-1} \{T = m\}\right)^c \in \mathcal{A}_{k-1}$ since $\{T = m\} \in \mathcal{A}_m \subseteq \mathcal{A}_{k-1}$ for $0 \leq m \leq k-1$, and cA_{k-1} is closed under finite unions and complements. Thus, $V_1, \ldots, V_N$ is a game system, and hence $\mathbb{E}[S_T] = \mathbb{E}[S_N^V] = 0$ by the previous theorem. ■

Suppose $T : \Omega \to \{0, 1, \dots, N\}$ is a stopping time. Then $\mathrm{Var}(S_T) = \mathbb{E}[S_T^2] - \mathbb{E}[S_T]^2 = \mathbb{E}[S_T^2]$. We can compute $\mathbb{E}[S_T^2]$ as follows; let $1 \le k \le N$. Then

$$S_k^2 = (S_{k-1} + X_k)^2 = S_{k-1}^2 + 2X_k S_{k-1} + 1. \tag{1.16}$$

Set $V_k = S_{k-1}\mathbf{1}_{\{T \ge k\}}$. Verify that $V_1, \dots, V_N$ is a game system. Then, we have

$$\begin{aligned} S_N^V &= \sum_{k=1}^N X_k V_k = \sum_{k=1}^N (S_{k-1}X_k)\mathbf{1}_{\{T\ge k\}} = \sum_{k=1}^N \frac{1}{2}(S_k^2 - S_{k-1}^2 - 1)\mathbf{1}_{\{T \ge k\}} \\ &= \frac{1}{2}\sum_{k=1}^N S_k^2 \mathbf{1}_{\{T\ge k\}} - \frac{1}{2}\sum_{k=1}^N S_{k-1}^2\mathbf{1}_{\{T\ge k\}} - \frac{1}{2}\sum_{k=1}^N \mathbf{1}_{\{T \ge k\}} \\ &= \frac{1}{2}\sum_{k=1}^N S_k^2\mathbf{1}_{\{T=k\}} - \frac{T}{2} = \frac{S_T^2 - T}{2}. \end{aligned} \tag{1.17}$$

By the previous theorem, we have $\mathbb{E}[S_N^V] = 0$, and hence $\mathbb{E}[S_T^2] = \mathbb{E}[T]$.

1.2.1 Ruin Problem

We have two players, one with capital a and the other with capital b. For $0 \le n \le N$, let S_n denote the gain of the first player. We define the first player's ruin as

$$\{\sigma_{-a} < \sigma_b,\, \sigma_{-a} \le N\}. \tag{1.18}$$

Define the ruin probability γ_N^1 to be the probability of this event, the first player's ruin. Similarly, we define the second player's ruin as

$$\{\sigma_b < \sigma_{-a},\, \sigma_b \le N\}, \tag{1.19}$$

and define the ruin probability γ_N^2 to be the probability of this event, the second player's ruin. Now define $T_N : \Omega \to \{0, 1, 2, \dots, N\}$ as $T_N = \min\{\sigma_{-a}, \sigma_b, N\}$. Then, T_N is a stopping time. It has been shown that $\mathbb{E}[S_{T_N}] = 0$. Explicitly writing out S_{T_N}, we have

$$S_{T_N} = \begin{cases} -a & \text{if } \sigma_{-a} < \sigma_b \text{ and } \sigma_{-a} \le N, \\ b & \text{if } \sigma_b < \sigma_{-a} \text{ and } \sigma_b \le N, \\ S_N & \text{if } \min(\sigma_{-a}, \sigma_b) \ge N. \end{cases} \tag{1.20}$$

Taking expectations, we have

$$\begin{aligned} &0 = -a\mathbb{P}(\sigma_{-a} < \sigma_b,\, \sigma_{-a} \le N) + b\mathbb{P}(\sigma_b < \sigma_{-a},\, \sigma_b \le N) + \mathbb{E}[S_N\mathbf{1}_{\{\min(\sigma_{-a},\sigma_b)\ge N\}}] \\ \implies\ &0 = -a\gamma_N^1 + b\gamma_N^2 + \mathbb{E}[S_N\mathbf{1}_{\{\min(\sigma_{-a},\sigma_b)\ge N\}}]. \end{aligned} \tag{1.21}$$

Here, we claim that $\mathbb{E}[S_N\mathbf{1}_{\{\min(\sigma_{-a},\sigma_b)\ge N\}}] \to 0$ as $N \to \infty$. To this end, first notice the bounds

$$0 \le |S_n|\,\mathbf{1}_{\{\min(\sigma_{-a},\sigma_b)\ge N\}} \le (b+a)\mathbf{1}_{\{\min(\sigma_{-a},\sigma_b)\ge N\}} \tag{1.22}$$

which then implies

$$0 \le \mathbb{E}[|S_n|\,\mathbf{1}_{\{\min(\sigma_{-a},\sigma_b)\ge N\}}] \le (b+a)\mathbb{P}(\min(\sigma_{-a},\sigma_b) \ge N) \le (b+a)c\frac{(b-a+1)}{\sqrt{N}} \to 0 \text{ as } N \to \infty. \tag{1.23}$$

We now make three 'leaps of faith;' they are

1. $\gamma^1 = \lim_{N\to\infty}\gamma_N^1$ and $\gamma^2 = \lim_{N\to\infty}\gamma_N^2$ exist.
2. $\gamma^1 + \gamma^2 = 1$.
3. $0 = -a\gamma^1 + b\gamma^2$.

The last two assumptions imply that $\gamma^1 = \frac{b}{a+b}$ and $\gamma^2 = \frac{a}{a+b}$.

1.3 Reflection Principle

Consider a random walk starting at the origin that ends up at $a+c$. Such a walk will have to pass through a at some point. We can then reflect the part of the walk after it hits a about the line $y = a$. This gives us a bijection between walks that end up at $a+c$ and walks that reach a and then end up at $a-c$. That is,

$$|\{\omega \in \Omega \mid S_n = a+c\}| = |\{\omega \in \Omega \mid S_n = a+c,\, \sigma_a \le n\}| = |\{\omega \in \Omega \mid S_n = a-c,\, \sigma_a \le n\}|. \tag{1.24}$$

This lemma immediately follows.

Lemma 1.7. $\mathbb{P}(S_n = a+c) = \mathbb{P}(\sigma_a \le n,\, S_n = a-c)$.

Theorem 1.8. $\mathbb{P}(\sigma_a \le n) = \mathbb{P}(S_n \notin [-a, a-1])$, *and* $\mathbb{P}(\sigma_a = n) = \frac{a}{n}\mathbb{P}(S_n = a)$.

Proof. We have

$$\begin{aligned}
\mathbb{P}(\sigma_a \le n) &= \sum_{b\in\mathbb{Z}} \mathbb{P}(\sigma_a \le n,\, S_n = b) = \sum_{b\in\mathbb{Z},\, b\ge a} \mathbb{P}(\sigma_a \le n,\, S_n = b) + \sum_{b\in\mathbb{Z},\, b<a} \mathbb{P}(\sigma_a \le n,\, S_n = b) \\
&= \mathbb{P}(S_n \ge a) + \sum_{b\in\mathbb{Z},\, b<a} \mathbb{P}(S_n = 2a-b) = \mathbb{P}(S_n \ge a) + \mathbb{P}(S_n > a) = \mathbb{P}(S_n \ge a) + \mathbb{P}(S_n < -a) \\
&= \mathbb{P}(S_n \notin [-a, a-1]).
\end{aligned} \tag{1.25}$$

On the other hand,

$$\begin{aligned}
\mathbb{P}(\sigma_a = n) &= \mathbb{P}(\sigma_a \le n) - \mathbb{P}(\sigma_a \le n-1) = \frac{1}{2}\left(\mathbb{P}(S_{n-1} = a-1) - \mathbb{P}(S_{n-1} = a+1)\right) \\
&= \frac{1}{2}\left(\binom{n-1}{\frac{n-1+a-1}{2}}\frac{1}{2^{n-1}} - \binom{n-1}{\frac{n-1-a+1}{2}}\frac{1}{2^{n-1}}\right) = \frac{1}{2^n}\cdot\frac{a}{n}\binom{n}{\frac{n+a}{2}} = \frac{a}{n}\mathbb{P}(S_n = a).
\end{aligned} \tag{1.26}$$

■

We now take as convention that $\sigma_0 = \min\{n \ge 1 \mid S_n = 0\}$ so that $\sigma_0 \ne 0$.

Lemma 1.9. $\mathbb{P}(\sigma_0 > 2n) = \mathbb{P}(S_{2n-1} = -1)$.

Proof. We have

$$\begin{aligned}
\mathbb{P}(\sigma_0 > 2n) &= \mathbb{P}(S_1 \ne 0, S_2 \ne 0, \dots, S_{2n} \ne 0) = 2\mathbb{P}(S_1 > 0, \dots, S_{2n} > 0) \\
&= 2\cdot\frac{1}{2^{2n}}\#\{\text{paths that start at 0 and don't hit } -1 \text{ in } 2n-1 \text{ steps}\} \\
&= \frac{1}{2^{2n-1}}\#\{\text{paths that start at 0 and stay below 1 in } 2n-1 \text{ steps}\} = \mathbb{P}(\sigma_1 > 2n-1) \\
&= 1 - \mathbb{P}(\sigma_1 \le 2n-1) = 1 - \mathbb{P}(S_{2n-1} \notin [0,1]) = 1 - \mathbb{P}(S_{2n-1} \ne -1) = \mathbb{P}(S_{2n-1} = -1).
\end{aligned} \tag{1.27}$$

■

1.3.1 Arcsine Law

August 5th.

The arcsine law is concerned with the analysis of the last visit to the origin till $2N$. That is, defining $L = \max\{0 \le n \le 2N \mid S_n = 0\}$, we wish to compute the order of L.

Theorem 1.10. *Let* $n \in \mathbb{N}_0$*, with* $n \le N$*. Then* $\mathbb{P}(L = 2n) = \frac{1}{2^{2N}}\binom{2n}{n}\binom{2N-2n}{N-n}$.

Remark 1.11. Suppose this theorem holds; using Stirling's formula, that $n! \sim n^n e^{-n}$ for large n, we can approximate $\mathbb{P}(L = 2n) \sim \frac{1}{\pi\sqrt{n(N-n)}} = \frac{1}{\pi N}\frac{1}{\sqrt{\frac{n}{N}(1-\frac{n}{N})}}$. The idea is to find the order of $\mathbb{P}(L \leq 2Nx) = \mathbb{P}(\frac{L}{2N} \leq x)$ for $x \in [0,1]$. Rewriting gives

$$\mathbb{P}(L \leq 2Nx) = \sum_{n \leq \lfloor Nx \rfloor} \mathbb{P}(L = 2n) = \sum_{n \leq \lfloor Nx \rfloor} \frac{1}{N} f(\frac{1}{N}) \tag{1.28}$$

where $f(x) = \frac{1}{\pi\sqrt{x(1-x)}}$. This is a Riemann sum, and hence we have $\mathbb{P}(L \leq 2Nx) \sim \int_0^x f(t)dt = \frac{2}{\pi}\arcsin(\sqrt{x})$.

Proof of theorem. We simply have

$$\mathbb{P}(L = 2n) = \mathbb{P}(S_{2n} = 0)\mathbb{P}(\sigma_0 > 2N - 2n) = \binom{2n}{n}\frac{1}{2^{2n}} \cdot \mathbb{P}(S_{2N-2n} = 0) = \frac{1}{2^{2N}}\binom{2n}{n}\binom{2N-2n}{N-n}. \tag{1.29}$$

■

1.4 Higher Dimensions and Infinite Walks

We are now fit to make sense of simple random walks on higher dimensions; let $\mathbb{Z}^d = \{(x_1, \ldots, x_d) \mid x_i \in \mathbb{Z}, 1 \leq i \leq d\}$ for $d \geq 1$. Also, let $|x| = \sqrt{\sum_{i=1}^d x_i^2}$. Then, for a fixed $N \geq 1$, we define

$$\Omega_N = \{\omega = (\omega_1, \ldots, \omega_N) \mid \omega_i \in \mathbb{Z}^d,\ |\omega_i| = 1,\ 1 \leq i \leq N\} \tag{1.30}$$

and $\xi_i : \Omega_N \to \mathbb{Z}^d$ as $\xi_i(\omega) = \omega_i$ for $1 \leq i \leq N$. We can then further define $X_n : \Omega_N \to \mathbb{Z}^d$ as $X_n(\omega) = \sum_{i=1}^n \xi_i(\omega)$ for $1 \leq n \leq N$. Moreover, $\mathcal{F} = \mathcal{P}(\Omega_N)$, and $\mathbb{P}(A) = \frac{|A|}{(2d)^N}$ for $A \in \mathcal{F}$. This is the simple symmetric random walk on $\mathbb{Z}^d$ of length N.

We wish to extend our notion of random walks for ones of infinite length; that is, the consistency of $\mathbb{P}_N$ as N varies. We define the projection map $\Pi_M : \Omega_N \to \Omega_M$ as $\omega \mapsto (\omega_1, \ldots, \omega_M) =: \tilde{\omega}$. Now fix $\tilde{\omega} \in \Omega_M$ so that

$$\mathbb{P}_N(\{\omega \in \Omega_N \mid \Pi_M(\omega) = \tilde{\omega}\}) = \frac{|\{\omega \in \Omega_N \mid \Pi_N(\omega) = \tilde{\omega}\}|}{(2d)^N} = \frac{(2d)^{N-M}}{(2d)^N} = (2d)^{-M} = \mathbb{P}_M(\{\tilde{\omega}\}). \tag{1.31}$$

Theorem 1.12 (*Kolmogorov consistency theorem*)**.** *There exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that $\Omega \equiv \Omega_\infty = \{(\omega_i)_{i\geq 1} \mid \omega_i \in \mathbb{Z}^d, |w_i| = 1\}$, with $\mathcal{F}$ suitabily defined, and $\mathbb{P} : \mathcal{F} \to [0,1]$ such that $\mathbb{P}(\omega \in \Omega \mid \Pi_M(\omega) = \tilde{\omega}) = \mathbb{P}_M(\{\tilde{\omega}\})$ for all $M \geq 1$ and $\tilde{\omega} \in \Omega_M$.*

A proof of this theorem is beyond the scope of this course; this lays the groundwork for the study of random walks of infinite length, by defining $\xi_i : \Omega \to \mathbb{Z}^d$ as $\xi_i(\omega) = \omega_i$ for $i \geq 1$, and $X_n : \Omega \to \mathbb{Z}^d$ as $X_n(\omega) = \sum_{i=1}^n \xi_i(\omega)$ for $n \geq 1$. This is the simple symmetric random walk on $\mathbb{Z}^d$ of infinite length.

1.4.1 Large Number Results

Let $\{X_n\}_{n\geq 1}$ be the simple symmetric random walk on $\mathbb{Z}^d$ of infinite length. Define

$$A = \{\omega \in \Omega \mid X_n(\omega)/n \to 0 \text{ as } n \to \infty\}. \tag{1.32}$$

Then, one can show that $\mathbb{P}(A) = 1$ despite $A \neq \Omega$. This is a consequence of *Stirling's law of large numbers*. Also,

$$\mathbb{P}\left(\frac{X_n}{\sqrt{n}} \leq x\right) \to \int_{-\infty}^x \frac{e^{-y^2/2}}{\sqrt{2\pi}} dy \text{ as } n \to \infty. \tag{1.33}$$

for when $d = 1$. This is a consequence of the *central limit theorem*.

Chapter 2

MARKOV CHAINS

For the simple symmetric random walk on $\mathbb{Z}^d$, we saw that when $\mathbb{P}(X_1 = i_1, \ldots, X_n = i_n)$ is non-zero, it is given by

$$\mathbb{P}(X_1 = i_1, \ldots, X_n = i_n) = \mathbb{P}(X_1 = i_1) \prod_{k=2}^{n} \mathbb{P}(X_k = i_k \mid X_{k-1} = i_{k-1}) = \prod_{k=1}^{n} \mathbb{P}(X_2 = i_k \mid X_1 = i_{k-1}) \tag{2.1}$$

where $i_0 = 0$. If the walk were not symmetric, we could then define

$$p_{ij} = \begin{cases} p & \text{if } i = j+1, \\ 1-p & \text{if } i = j-1, \\ 0 & \text{otherwise.} \end{cases} \tag{2.2}$$

With this, the above probability is $\prod_{k=1}^{n} p_{i_k i_{k-1}}$. In fact, the walk need not simple too, with steps of arbitrary size taking place. Moreover, $\mathbb{Z}^d$ can be replaced by any discrete space S. This is where we continue.

2.1 Discrete Time and Space Markov Chain

Let S be any countable set, called our state space, and X_n denote the state of the system at time n. The dynamics of the system is given by the transition probabilities; X_n moves from $i \in S$ to $j \in S$ with probability p_{ij}, which depends only on i and j. Some obvious properties include $p_{ij} \in [0, 1]$ and $\sum_{j \in S} p_{ij} = 1$. The transition probabilities can be arranged in a matrix $P = [p_{ij}]_{i,j \in S}$, called the one-step *transition matrix*.

Definition 2.1. Let μ be any probability on $(S, \mathcal{P}(S))$, and let $P = [p_{ij}]_{i,j \in S}$ be a one-step transition matrix. Moreover, let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with $(X_n)_{n \geq 0}$ a sequence of S-valued random variables whose joint distribution is given by

$$\mathbb{P}(X_0 = i_0, X_1 = i_1, \ldots, X_n = i_n) = \mu(\{i_0\}) \prod_{k=1}^{n} p_{i_{k-1} i_k}. \tag{2.3}$$

Then $\{X_n\}_{n \geq 0}$ is called a *Markov chain* on S with initial distribution μ and transition matrix P.

To find the distribution of X_n, we can use the law of total probability to get

$$\mathbb{P}(X_n = j) = \sum_{i_0, \ldots, i_{n-1} \in S} \mathbb{P}(X_0 = i_0, \ldots, X_{n-1} = i_{n-1}, X_n = j) = \sum_{i_0, \ldots, i_{n-1} \in S} \mu(\{i_0\}) \prod_{k=1}^{n} p_{i_{k-1} i_k}. \tag{2.4}$$

We can also show that the distribution of X_n when the previous $n-1$ steps are known, depends only on the last step; we have

$$\begin{aligned}\mathbb{P}(X_n = j \mid X_{n-1} = i_{n-1}, \dots, X_0 = i_0) &= \frac{\mathbb{P}(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = j)}{\mathbb{P}(X_0 = i_0, \dots, X_{n-1} = i_{n-1})} \\ &= \frac{\mu(\{i_0\}) \prod_{k=1}^{n} p_{i_{k-1}i_k}}{\mu(\{i_0\}) \prod_{k=1}^{n-1} p_{i_{k-1}i_k}} = p_{i_{n-1}j}. \end{aligned} \tag{2.5}$$

On the other hand,

$$\begin{aligned}\mathbb{P}(X_n = j \mid X_{n-1} = i_{n-1}) &= \frac{\mathbb{P}(X_n = j, X_{n-1} = i_{n-1})}{\mathbb{P}(X_{n-1} = i_{n-1})} \\ &= \frac{\sum_{i_0,\dots,i_{n-2}\in S} \mu(\{i_0\}) \prod_{k=1}^{n} p_{i_{k-1}i_k} p_{i_{n-1}j}}{\sum_{i_0,\dots,i_{n-2}\in S} \mu(\{i_0\}) \prod_{k=1}^{n} p_{i_{k-1}i_k}} = p_{i_{n-1}j}. \end{aligned} \tag{2.6}$$

Thus, the distribution of X_n depends only on the previous state, and not on the entire history. Also, the distribution of X_n given the start state $X_0 = i_0$ is

$$\mathbb{P}(X_n = j \mid X_0 = i_0) = \sum_{i_1,\dots,i_{n-1}\in S} \prod_{k=1}^{n} p_{i_{k-1}i_k}. \tag{2.7}$$

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If, in the distribution of X_n, the starting point $X_0 = i$ is specified, we can write the distribution as $\mathbb{P}(X_n = j \mid X_0 = i) = p_{ij}^n$, where p_{ij}^n is the (i,j)-th entry of the n-step transition matrix P^n. Further, if S is finite, then whenever $r + s = n$, $P^n = P^r P^s$, and thus $p_{ij}^n = \sum_{k\in S} p_{ik}^r p_{kj}^s$. Expanding gives

$$\mathbb{P}(X_n = j \mid X_0 = i) = \sum_{k\in S} \mathbb{P}(X_r = k \mid X_0 = i) \cdot \mathbb{P}(X_s = j \mid X_0 = k). \tag{2.8}$$

This is called the *Chapman-Kolmogorov equation*.

Example 2.2. Let S be a countable vertex set, and let $E \subseteq \{\{i,j\} \mid i,j \in S\}$ be a set of unordered edges. Let $\eta : E \to (0,\infty)$ be a function that assigns a positive weight to each edge. We define $\eta_{ij} = \eta(\{i,j\})$ for all $\{i,j\} \in E$, and $\eta_{ij} = 0$ if $\{i,j\} \notin E$. Then further define $\eta_i = \sum_{j\in S} \eta_{ij}$. We may assume that $\eta_i < \infty$ for all $i \in S$. Now, set $p_{ij} = \eta_{ij}/\eta_i$ for all $i,j \in S$. Then, $0 \le p_{ij} \le 1$ and $\sum_{j\in S} p_{ij} = 1$. Thus, $P = [p_{ij}]_{i,j\in S}$ is a transition matrix. The Markov chain with this transition matrix, and S as the state space, is simply a random walk on the weighted graph (S,E,η). In particular, if $\eta_{ij} = 1$ for all $\{i,j\} \in E$, then $p_{ij} = 1/\deg(i)$, where $\deg(i)$ is the degree of vertex i. This the simple random walk on the graph (S,E).

Example 2.3. Let $S = \{0,1\}$, and let $P = \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix}$ with $0 \le \alpha, \beta \le 1$. Also define the initial distribution as $\mu(\{0\}) = \mu_0$ and $\mu(\{1\}) = \mu_1 = 1 - \mu_0$. Then, $\{X_n\}_{n\ge 0}$ is a Markov chain on S with transition matrix P and initial distribution μ. This is called a two-state Markov chain.

Let us expand on the above example; the distribution of X_n is can be given as follows. We know $P^0 = I$ and $P^n = P^{n-1}P$ for $n \ge 1$. Thus, we can write

$$p_{00}^n = p_{00}^{n-1}(1-\alpha) + p_{01}^{n-1}\beta = p_{00}^{n-1}(1-\alpha-\beta) + \beta. \tag{2.9}$$

This is a linear recurrence relation; first checking for constant solutions $p_{00}^n \equiv p$ gives $p = \frac{\beta}{\alpha+\beta}$. For a more general solution, set $x_n = p_{00}^n - \frac{\beta}{\alpha+\beta}$, which gives $x_n = (1-\alpha-\beta)x_{n-1}$. Thus, $x_n = (1-\alpha-\beta)^n x_0$. With $x_0 = p_{00}^0 - \frac{\beta}{\alpha+\beta} = 1 - \frac{\beta}{\alpha+\beta} = \frac{\alpha}{\alpha+\beta}$, we have

$$p_{00}^n = (1-\alpha-\beta)^n \frac{\alpha}{\alpha+\beta} + \frac{\beta}{\alpha+\beta}. \tag{2.10}$$

One can also show that

$$p_{10}^n = \frac{\beta}{\alpha+\beta} - \frac{\beta}{\alpha+\beta}(1-\alpha-\beta)^n. \tag{2.11}$$

Thus, the distribution of X_n is given by

$$\begin{aligned}\mathbb{P}(X_n = 0) &= \mathbb{P}(X_n = 0 \mid X_0 = 0)\cdot\mathbb{P}(X_0 = 0) + \mathbb{P}(X_n = 0 \mid X_0 = 1)\cdot\mathbb{P}(X_0 = 1) = p_{00}^n\mu_0 + p_{10}^n\mu_1 \\ &= (p_{00}^n - p_{10}^n)\mu_0 + p_{10}^n = \frac{\beta}{\alpha+\beta} + (1-\alpha-\beta)^n\left(\mu_0 - \frac{\beta}{\alpha+\beta}\right), \end{aligned} \tag{2.12}$$

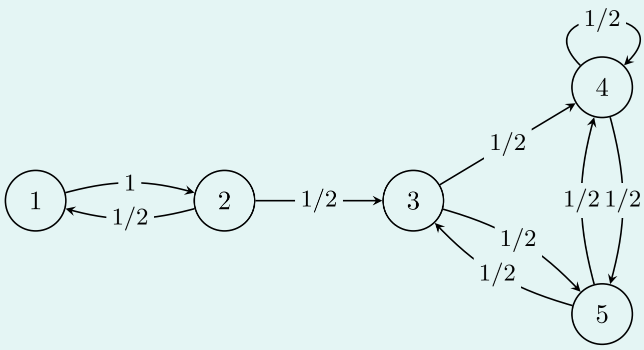
and

$$\mathbb{P}(X_n = 1) = 1 - \mathbb{P}(X_n = 0) = \frac{\alpha}{\alpha+\beta} - (1-\alpha-\beta)^n\left(\mu_1 - \frac{\alpha}{\alpha+\beta}\right). \tag{2.13}$$

Immediately, we can make some observations.

1. If we take $\mu_0 = \frac{\beta}{\alpha+\beta}$, then $\mathbb{P}(X_n = 0) = \frac{\beta}{\alpha+\beta}$ and $\mathbb{P}(X_n = 1) = \frac{\alpha}{\alpha+\beta}$ for all n. Thus, the distribution of X_n is independent of n.
2. If $\alpha+\beta > 0$, or $1-\alpha-\beta < 1$, then $\lim_{n\to\infty}(1-\alpha-\beta)^n = 0$. Thus, $\lim_{n\to\infty}\mathbb{P}(X_n = 0) = \frac{\beta}{\alpha+\beta}$ and $\lim_{n\to\infty}\mathbb{P}(X_n = 1) = \frac{\alpha}{\alpha+\beta}$.

Example 2.4. Consider the following weighted digraph.



Intuitively, it is clear that the Markov chain will spend a 'finite' amount of time in states 1 and 2, and an 'infinite' amount of time in states 3, 4, and 5. We can formalize this by computing the distribution of X_n as $n \to \infty$. We wish to show that $p_{11}^n \to 0$ as $n \to \infty$.

2.1.1 Recurrence and Transience

Let $\{X_n\}_{n\geq 0}$ be a Markov chain on a countable state space S with transition matrix $P = [p_{ij}]_{i,j\in S}$, and initial distribution μ. For all $n \geq 1$ and $i \in S$, define

$$N_n(i) = \sum_{k=0}^{n} \mathbf{1}_{\{X_k = i\}} \quad \text{and} \quad N(i) = \sum_{k=0}^{\infty} \mathbf{1}_{\{X_k = i\}}. \tag{2.14}$$

That is, $N_n(i)$ is the number of visits to state i up to time n, and $N(i)$ is the total number of visits to state i. From here, we can define the following.

Definition 2.5. A state $i \in S$ is called *recurrent* if $\mathbb{P}(N(i) = \infty) = 1$, and *transient* if $\mathbb{P}(N(i) < \infty) = 1$. The chain $\{X_n\}_{n\geq 1}$ is called recurrent if all states are recurrent, and transient if all states are transient.

We shall later show that every state in a Markov chain is either recurrent or transient. For $i \in S$, let $T_i = \min\{n \geq 1 \mid X_n = i\}$ be the first return time to state i.

Lemma 2.6. *Let $i \in S$, and $m \geq 1$. Then*

$$\mathbb{P}(N(i) \geq m \mid X_0 = i) = \mathbb{P}(T_i < \infty \mid X_0 = i)^m. \tag{2.15}$$

Proof. We have

$$\{N(i) \geq m\} = \bigcup_{k=1}^{\infty} \{N(i) \geq m, T_i = k\}. \tag{2.16}$$

Suppose, for $k \geq 1$, that $\mathbb{P}(T_i = k \mid X_0 = i) > 0$. Then, by the Markov property, we have

$$\begin{aligned} \mathbb{P}(N(i) \geq m, T_i = k \mid X_0 = i) &= \mathbb{P}(N(i) \geq m \mid T_i = k, X_0 = i) \cdot \mathbb{P}(T_i = k \mid X_0 = i) \\ &= \mathbb{P}(N(i) \geq m - 1 \mid X_0 = i) \cdot \mathbb{P}(T_i = k \mid X_0 = i). \end{aligned} \tag{2.17}$$

Note that this equation is trivially true if $\mathbb{P}(T_i = k \mid X_0 = i) = 0$. Thus, we have

$$\begin{aligned} \mathbb{P}(N(i) \geq m \mid X_0 = i) &= \sum_{k=1}^{\infty} \mathbb{P}(N(i) \geq m, T_i = k \mid X_0 = i) \\ &= \sum_{k=1}^{\infty} \mathbb{P}(N(i) \geq m - 1 \mid X_0 = i) \cdot \mathbb{P}(T_i = k \mid X_0 = i) \\ &= \mathbb{P}(N(i) \geq m - 1 \mid X_0 = i) \cdot \sum_{k=1}^{\infty} \mathbb{P}(T_i = k \mid X_0 = i) \\ &= \mathbb{P}(N(i) \geq m - 1 \mid X_0 = i) \cdot \mathbb{P}(T_i < \infty \mid X_0 = i). \end{aligned} \tag{2.18}$$

Inductively, we conclude that

$$\mathbb{P}(N(i) \geq m \mid X_0 = i) = \mathbb{P}(T_i < \infty \mid X_0 = i)^m. \tag{2.19}$$

■

Proposition 2.7. *Let $i \in S$. Then i is recurrent if and only if $\mathbb{P}(T_i < \infty \mid X_0 = i) = 1$, and transient if and only if $\mathbb{P}(T_i < \infty \mid X_0 = i) < 1$.*

Immediately from here, a state must be recurrent or transient.

Proof. Note that $\{N(i) = \infty\} = \bigcap_{m=1}^{\infty} \{N(i) \geq m\}$. Thus, by continuity of probability measures, we have

$$\mathbb{P}(N(i) = \infty) = \lim_{m \downarrow \infty} \mathbb{P}(N(i) \geq m). \tag{2.20}$$

Therefore,

$$\mathbb{P}(N(i) = \infty \mid X_0 = i) = \lim_{m \to \infty} \mathbb{P}(N(i) \geq m \mid X_0 = i) = \lim_{m \to \infty} \mathbb{P}(T_i < \infty \mid X_0 = i)^m \tag{2.21}$$

which is 1 if $\mathbb{P}(T_i < \infty \mid X_0 = i) = 1$, and 0 if $\mathbb{P}(T_i < \infty \mid X_0 = i) < 1$, by the previous lemma. Thus, i is recurrent if and only if $\mathbb{P}(T_i < \infty \mid X_0 = i) = 1$, and transient if and only if $\mathbb{P}(T_i < \infty \mid X_0 = i) < 1$. ■

Proposition 2.8. *Let $i \in S$. Then i is transient if and only if $\sum_{n=0}^{\infty} p_{ii}^n < \infty$, and recurrent if and only if $\sum_{n=0}^{\infty} p_{ii}^n = \infty$.*

Proof. Note that $\mathbb{P}(T_i < \infty \mid X_0 = i) < 1$ if and only if $\sum_{m=1}^{\infty} \mathbb{P}(T_i < \infty \mid X_0 = i)^m < \infty$, which is equivalent to $\sum_{m=1}^{\infty} \mathbb{P}(N(i) \geq m - 1 \mid X_0 = i) < \infty$. But, $\sum_{m=1}^{\infty} \mathbb{P}(N(i) \geq m - 1 \mid X_0 = i) = \sum_{m=0}^{\infty} \mathbb{P}(N(i) \geq m \mid X_0 = i) = \mathbb{E}[N(i) \mid X_0 = i]$. This last term can be rewritten as

$$\mathbb{E}[N(i) \mid X_0 = i] = \mathbb{E}\left[\sum_{n=1}^{\infty} \mathbf{1}_{\{X_n = i\}} \,\middle|\, X_0 = i\right] = \sum_{n=1}^{\infty} \mathbb{E}[\mathbf{1}_{\{X_n = i\}} \mid X_0 = i] = \sum_{n=1}^{\infty} p_{ii}^n. \tag{2.22}$$

■

Example 2.9. Consider the state space $S = \mathbb{Z}$ with $p_{ij} = 1/2$ if $j = i+1$ or $j = i-1$, and $p_{ij} = 0$ otherwise. This is the simple symmetric random walk on $\mathbb{Z}$. We claim that $0 \in S$ is recurrent. To see this, we have

$$p_{00}^{2n} = \frac{1}{2^{2n}}\binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}} \implies \sum_{n=0}^{\infty} p_{00}^{2n} = \infty. \tag{2.23}$$

Therefore, 0 is recurrent. Now, let $i \in \mathbb{Z}$ be any state. There exist $r, s \in \mathbb{N}$ such that $p_{i0}^r > 0$ and $p_{0i}^s > 0$. If we let $n \in \mathbb{N}$, then $n = r + s + k$ for some $k \geq 1$. Thus, $p_{ii}^n \geq p_{i0}^r p_{00}^k p_{0i}^s$. Therefore, $\sum_{n=0}^{\infty} p_{ii}^n \geq p_{i0}^r p_{0i}^s \sum_{k=0}^{\infty} p_{00}^k = \infty$, and thus i is recurrent. Since i was arbitrary, the chain is recurrent.

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